

# Rotating Away the Location Degeneracy: An Exact Reparameterization for Hamiltonian Monte Carlo in Hierarchical and Joint Models

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## Abstract

In regression models with subject-level random effects, the likelihood identifies the sum of a subject-constant fixed effect and the matching combination of random effects, but not its split. Under the non-centered parameterization that Hamiltonian Monte Carlo (HMC) software uses by default, only the prior locates the split, and the resulting ridge can slow sampling of the reported coefficients. We characterize every random-effect direction the fixed effects can absorb, and the condition a survival submodel adds in joint longitudinal-survival models. One fixed orthogonal rotation of the standardized random effects gathers these directions into a small block of sampling coordinates. It is exact for any random-effect covariance, and a dense mass-matrix block over this block and the fixed effects largely whitens the ridge. The effective sample size per second of the intercept and time slope rose 14- and 21-fold in simulation (geometric means), and that of the intercept and subject-constant coefficients 17- to 28-fold on two joint-model datasets. For the association parameter of a joint model, an ideal two-block sampler's lag-1 autocorrelation equals the share of its posterior variance explained by the other parameters, which on real data was at least 0.85. Supplementary materials for this article are available online.

*Keywords:* Identifiability; Longitudinal and time-to-event data; Mixed models; No-U-Turn sampler; Non-centered parameterization; Preconditioning.

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# 1 Introduction

Mixed models with subject-level random effects are routinely fitted by Hamiltonian Monte Carlo (HMC) and its adaptive variant, the No-U-Turn sampler (NUTS; [Hoffman & Gelman 2014](#)), in Stan ([Carpenter et al. 2017](#)), brms ([Bürkner 2017](#)), rstanarm and NumPyro ([Phan et al. 2019](#)). The standard choice, and the default in brms and rstanarm, is the non-centered parameterization  $b = B_{\text{std}}L^\top$ , with  $B_{\text{std}}$  standard normal and  $L$  a Cholesky factor of the random-effect covariance. It removes the funnel between the random effects and their scale ([Neal 2003](#), [Betancourt & Girolami 2015](#)).

It leaves a second, simpler problem in place. Suppose the fixed-effects design contains a column that is constant within subject: the intercept, a treatment arm, sex. Adding a constant  $c$  to every subject’s random intercept and subtracting  $c$  from the fixed intercept leaves every fitted value unchanged, so only the priors pin the split, at scale  $\sigma_0/\sqrt{N}$  for  $N$  subjects when the fixed intercept’s prior is diffuse. The same holds for every subject-constant covariate, whose coefficient trades off against a covariate-weighted combination of the random intercepts, and, with a random slope on time, for the fixed time slope and the mean random slope. In HMC’s sampling coordinates these directions are spread across all  $N$  subjects, the posterior has a ridge along them, and the sampler can mix slowly on the intercept, the subject-constant coefficients and the time slope: the coefficients an analyst reports. In a joint model of the `pbc2` data, for instance, the posterior correlation between the fixed intercept and the mean random intercept was  $-0.97$  (Section [2.2](#)).

The problem is well understood for Gibbs samplers. Hierarchical centering ([Gelfand et al. 1995](#)) removes it at the price of the funnel, and the choice between centered and non-centered coordinates has a large literature ([Papaspiliopoulos et al. 2003, 2007](#), [Yu & Meng 2011](#)). Other approaches remove the directions: sum-to-zero constraints, whose effect on Gibbs and NUTS is quantified by [Zanella & Roberts \(2021\)](#); restricted spatial regres-

sion (Reich et al. 2006, Hodges & Reich 2010); and, exactly, the transformation of Vines et al. (1996), which integrates the grand-mean directions out. Yang & Liu (2021) use the covariate-weighted direction to analyze Gibbs samplers, and Browne et al. (2009) name the problem, suggesting that predictors be made orthogonal to the cluster indicators.

This paper keeps the absorbable directions and makes them sampling coordinates. We construct an orthonormal basis for every direction the fixed effects can absorb, in every random-effect column, and apply one fixed rotation to the standardized random effects so that these directions become a small block  $U$  of  $k \times q$  coordinates, with  $k$  directions and  $q$  random-effect columns. A single dense block of the HMC mass matrix over  $(\beta, U)$  then largely whitens the ridge. The rotation acts on the subject index and the covariance factor on the column index, so they commute: the construction is exact for every value of the random-effect covariance, with no constraint, no correction and no change to any reported quantity.

The motivating application is the Bayesian joint model of a longitudinal marker and a time-to-event outcome (Wulfsohn & Tsiatis 1997, Henderson et al. 2000, Tsiatis & Davidian 2004, Rizopoulos 2012), as fitted with NUTS by the R package `jmj` (Guo 2026). There the survival submodel adds a condition on which directions remain absorbable, and a second question arises. On the association parameter  $\alpha$ , the primary estimand, NUTS gave 3 to 8 times the effective sample size (ESS) per second of the Metropolis-within-Gibbs sampler of JMbays2 (Rizopoulos et al. 2026) without any reparameterization. We explain most of the per-draw part of that gap, on two datasets where we measure the relevant quantity, with an ideal two-block sampler, whose lag-1 autocorrelation for  $\alpha$  equals the share of  $\alpha$ 's posterior variance that the other parameters explain (Liu et al. 1994).

Section 2 sets out the model and the degeneracy. Section 3 characterizes the absorbable directions for any design, including one where a natural column-by-column check misses the

slope direction, and the condition a survival submodel imposes. Section 4 gives the rotation, which costs  $O(Nkq)$  per gradient, and an analysis of the ridge that explains why a diagonal metric cannot handle it, why one dense block can, and where a fixed metric is weakest. Section 5 treats  $\alpha$ . Sections 6–8 give evidence from simulation, two mixed models and two joint-model datasets, including a comparison with JMbays2, and Section 9 discusses related work and limits. Section S6 of the Supplementary Material lists the script that reproduces each result.

## 2 Model and the location degeneracy

### 2.1 Mixed models and joint models

Index subjects by  $i = 1, \dots, N$  and subject  $i$ 's longitudinal observations by  $j = 1, \dots, n_i$ , taken at times  $t_{ij}$ . Subject  $i$  has a fixed-effects design  $X_i \in \mathbb{R}^{n_i \times p}$  and a random-effects design  $Z_i \in \mathbb{R}^{n_i \times q}$ , and the linear predictor is

$$\eta_{ij} = x_{ij}^\top \beta + z_{ij}^\top b_i, \quad b_i \sim \mathcal{N}_q(0, \Sigma), \quad \Sigma = LL^\top, \quad (1)$$

with  $L = \text{diag}(\sigma_b) L_{\text{corr}}$  lower triangular; for a random intercept and slope,  $\sigma_b = (\sigma_0, \sigma_1)$  and  $\rho$  is their correlation. The response is Gaussian with mean  $\eta_{ij}$  and variance  $\sigma_e^2$ , or follows a generalized linear model with linear predictor  $\eta_{ij}$ . Collect the random effects as the rows of  $B \in \mathbb{R}^{N \times q}$ . The non-centered parameterization samples

$$B_{\text{std}} \in \mathbb{R}^{N \times q}, \quad \text{entries iid } \mathcal{N}(0, 1), \quad B = B_{\text{std}} L^\top, \quad (2)$$

and  $\theta$  collects every other parameter, including  $\beta$ ,  $\sigma_b$  and  $L_{\text{corr}}$ .

In a joint longitudinal-survival model the longitudinal submodel is Gaussian, and subject  $i$ 's current underlying marker value

$$m_i(t) = x_i(t)^\top \beta + z_i(t)^\top b_i \quad (3)$$

enters a proportional hazards model,

$$h_i(t) = h_0(t) \exp\{\gamma^\top w_i + \alpha m_i(t)\}, \quad (4)$$

with baseline covariates  $w_i$  and association parameter  $\alpha$ , where  $x_i(t)$  and  $z_i(t)$  are the design rows at time  $t$ . In the `jmj` fits reported here the log baseline hazard is a cubic B-spline with five interior knots and a second-order random-walk prior (the package’s penalized option; its default puts independent normal priors on the coefficients), and the cumulative hazard is computed by Gauss–Legendre quadrature, so the random effects enter the likelihood through  $\eta_{ij}$  and through  $m_i(t)$  at each event time and quadrature node. The priors on  $\beta$ ,  $\gamma$  and  $\alpha$  are normal, those on the random-effect standard deviations gamma or half-normal, and that on  $L_{\text{corr}}$  LKJ (Lewandowski et al. 2009); nothing that follows depends on these choices beyond the iid standard normal prior on  $B_{\text{std}}$ .

## 2.2 The degeneracy

Take a random intercept and slope,  $z_{ij} = (1, t_{ij})^\top$ , and a fixed-effects design with an intercept, a time column and subject-constant covariates such as age and sex. Three kinds of shift leave every  $\eta_{ij}$  and every  $m_i(t)$  unchanged:

- $b_{i0} \mapsto b_{i0} + c$  for all  $i$ , with  $\beta_0 \mapsto \beta_0 - c$ ;
- $b_{i0} \mapsto b_{i0} + c \text{ age}_i$  for all  $i$ , with  $\beta_{\text{age}} \mapsto \beta_{\text{age}} - c$ , and likewise for every subject-constant covariate;
- $b_{i1} \mapsto b_{i1} + c$  for all  $i$ , with  $\beta_{\text{time}} \mapsto \beta_{\text{time}} - c$ .

The second kind adds one direction for each subject-constant covariate; the third is present whenever there is a random slope. Along each shift the likelihood is exactly invariant, and only the priors locate the split. For the mean intercept direction, with independent random intercepts and a diffuse prior on  $\beta_0$ , the scale is  $\sigma_0/\sqrt{N}$ ; a proper prior on  $\beta_0$  narrows it

further. We use “degeneracy” for this likelihood invariance. The posterior is proper and the split is identified, weakly: the posterior has a ridge, not a flat direction.

In the coordinates HMC samples, a shift of this kind moves all  $N$  entries of one or both columns of  $B_{\text{std}}$  together (for the mean intercept direction with uncorrelated random effects, each by  $c/\sigma_0$ ), against one or a few coordinates of  $\beta$ . With a diagonal metric, the default adaptation in Stan and NumPyro, the kinetic energy rescales coordinates one at a time and cannot represent the covariance along such a direction; a dense metric can, but over all  $Nq + p$  coordinates (Section 4). In a fit of the `pbc2` data (Section 8) with an intercept and one subject-constant covariate, the posterior correlation between  $\beta_0$  and the mean random intercept was  $-0.971$ , and between the covariate’s coefficient and the matching covariate-weighted combination  $-0.972$ . The two coefficients had minimum effective sample sizes of 441 and 526 from 4000 draws, against 2893 and more than 4000 for the identified sums. Stan’s user’s guide documents the same mechanism for two intercepts under weak priors (Stan Development Team 2026).

The degeneracy is not a funnel, whose curvature changes with a scale parameter so that no single metric fits everywhere. With the variance components held fixed, the posterior of  $\beta$  and the random effects is Gaussian in the linear model and the curvature along the ridge is constant: the ridge is narrow in a direction a diagonal metric cannot represent, which a fixed Euclidean metric can largely correct once the direction is exposed (Section 4.4 gives what remains when the variance components vary).

### 3 Absorbable directions

The rotation of Section 4 is exact for any fixed orthogonal matrix; whether it helps depends on the basis it is built from. The basis should contain every direction the fixed effects can absorb, and only such directions. This section characterizes them.

Fix a random-effect column  $c$  and write  $z_i = Z_i e_c \in \mathbb{R}^{n_i}$  for subject  $i$ 's entries of it; in this section only,  $z_i(t)$  denotes the scalar entry of column  $c$  in the design row at time  $t$ , rather than the whole row of (3).

**Definition 1** (Absorbable direction). A vector  $v \in \mathbb{R}^N$  is absorbable for column  $c$  if there is a witness  $d \in \mathbb{R}^p$  such that the shift  $Be_c \mapsto Be_c + v$ ,  $\beta \mapsto \beta + d$  leaves every linear predictor and every marker trajectory, and hence the likelihood, unchanged:

$$X_i d + v_i z_i = 0 \quad \text{for every } i, \tag{L}$$

$$x_i(t)^\top d + v_i z_i(t) = 0 \quad \text{for every } i \text{ and every } t \in [0, \tau]. \tag{T}$$

Write  $\mathcal{A}_c$  for the set of absorbable directions and  $\mathcal{A}_c^{\text{long}}$  for the set satisfying (L) alone, which is the whole requirement in a mixed model without a survival submodel.

Both sets are linear subspaces of  $\mathbb{R}^N$ . The likelihood uses  $m_i(t)$  only up to each subject's event or censoring time, so requiring (T) on all of  $[0, \tau]$  is slightly more than likelihood invariance needs; it makes  $\mathcal{A}_c$  a structural property of the model rather than of the observed follow-up. It is sufficient for likelihood invariance, not necessarily the largest set of shifts that leaves a particular observed likelihood unchanged. The prior on the random effects is not invariant under these shifts, and that asymmetry is what identifies the split at all.

### 3.1 Characterization and construction

Let  $\mathcal{T} : \mathbb{R}^p \rightarrow \bigoplus_i \mathbb{R}^{n_i}$ ,  $(\mathcal{T}d)_i = X_i d$ , map a shift of  $\beta$  to its effect on the stacked linear predictor, and  $\mathcal{U}_c : \mathbb{R}^N \rightarrow \bigoplus_i \mathbb{R}^{n_i}$ ,  $(\mathcal{U}_c v)_i = v_i z_i$ , do the same for a shift of random-effect column  $c$ . Let  $P_{\mathcal{T}}$  be the orthogonal projector onto range  $\mathcal{T}$  and, for  $z_i \neq 0$ ,  $P_{z_i}$  the projector onto  $\text{span}(z_i)$ . Let  $S_c = \{i : z_i \neq 0\}$  be the subjects with at least one observation at which column  $c$  is nonzero, and let  $M_c \in \mathbb{R}^{(\sum_i n_i) \times p}$  stack  $(I - P_{z_i})X_i$  for  $i \in S_c$  and  $X_i$  for  $i \notin S_c$ .

**Proposition 1** (Characterization). (a)  $\mathcal{A}_c^{\text{long}} = \{v : \mathcal{U}_c v \in \text{range } \mathcal{T}\}$ , the preimage of

range  $\mathcal{T}$  under  $\mathcal{U}_c$ , and equals  $\ker\{(I - P_{\mathcal{T}})\mathcal{U}_c\}$ . It depends on the fixed-effects design only through its column space.

(b) A vector  $d$  witnesses some  $v \in \mathcal{A}_c^{\text{long}}$  if and only if  $d \in \ker M_c$ . The witnessed  $v$  then has  $v_i = -z_i^\top X_i d / z_i^\top z_i$  for  $i \in S_c$ , and  $v_i$  is unrestricted for  $i \notin S_c$ .

*Proof.* (a) Condition (L) says  $\mathcal{T}d = -\mathcal{U}_c v$  for some  $d$ , that is,  $\mathcal{U}_c v \in \text{range } \mathcal{T}$ , which is equivalent to  $(I - P_{\mathcal{T}})\mathcal{U}_c v = 0$ . (b) For  $i \in S_c$ ,  $X_i d = -v_i z_i$  holds for some scalar  $v_i$  if and only if  $X_i d \in \text{span}(z_i)$ , that is,  $(I - P_{z_i})X_i d = 0$ , and then  $v_i = -z_i^\top X_i d / z_i^\top z_i$ . For  $i \notin S_c$ , (L) reads  $X_i d = 0$  whatever  $v_i$  is.  $\square$

Part (b) is the construction used in practice. In a mixed model without a survival submodel, a subject with  $z_i = 0$  has no random effect in column  $c$  in the likelihood, so we may take  $S_c$  to be every subject. One singular value decomposition of the  $(\sum_i n_i) \times p$  matrix  $M_c$  gives a basis of  $\ker M_c$ , each basis vector  $d$  gives one direction  $v(d)$ , and, if  $X$  has full column rank,  $d \mapsto v(d)$  is injective, so  $\dim \mathcal{A}_c^{\text{long}} = \dim \ker M_c \leq p$ . The computation is done once, before sampling. In the implementation every returned direction is also checked against (L) before use, so a loose rank tolerance cannot admit a direction that is not absorbable.

## 3.2 The survival submodel

In a joint model the requirement (T) quantifies over all  $t \in [0, \tau]$ , while  $M_c$  sees only the observation times.

**Condition S** (Structural extension). For every  $d \in \ker M_c$  and every subject  $i$  there is a scalar  $v_i(d)$ , equal to  $-z_i^\top X_i d / z_i^\top z_i$  when  $i \in S_c$ , such that  $x_i(t)^\top d + v_i(d) z_i(t) = 0$  for every  $t \in [0, \tau]$ . (Under the assumption of Proposition 2,  $v_i(d)$  is then unique for every  $i$ .)

**Proposition 2** (Joint model). Suppose every subject  $i \notin S_c$  has some  $t \in [0, \tau]$  with  $z_i(t) \neq 0$ . Under Condition S,  $\mathcal{A}_c = \{v(d) : d \in \ker M_c\}$ . For a subject  $i \notin S_c$ ,  $v_i(d)$

is determined by (T) at any  $t$  with  $z_i(t) \neq 0$ ; in particular it is not free, although the longitudinal data alone leave it free.

The proof is in Appendix A. Condition S holds for subject-constant columns (absorbed by the intercept column  $z_i(t) \equiv 1$ ), the time column (absorbed by a random slope  $z_i(t) = t$ ) and covariate-by-time interactions: in each, the identity holds as a function of  $t$ , not by coincidence at the observed times. More generally it holds when the functions  $t \mapsto x_i(t)^\top d$  and  $t \mapsto z_i(t)$  are known to lie in a finite-dimensional function space determined by its values at the subject’s distinct observation times, such as polynomials of degree below  $n_i$  (Appendix A). For spline bases it can fail for subjects with few observations relative to the flexibility of the basis. It is cheap to check: `jmjax` evaluates the identity at the event times and quadrature nodes at every fit and warns if the worst relative residual exceeds  $10^{-6}$ . Proposition 2 also settles the entry of a subject with no observation at which the column is nonzero, such as a baseline-only subject for the slope column: it is forced by the survival submodel, and setting it to zero would produce a direction the likelihood does identify.

### 3.3 A design where the natural check fails

A natural construction, and the one `jmjax` originally used, looks for columns of  $X$  that are *individually* subject-constant multiples of  $z$ : column  $j$  is kept if  $X_i e_j = s_i z_i$  for every  $i$ , and the basis is the span of the resulting vectors  $s$ . This finds a subspace of  $\mathcal{A}_c^{\text{long}}$ , but not always all of it.

**Example 1.** Let every subject be observed at the same  $m \geq 3$  distinct times, let  $Z_i = [1, t]$  with  $c$  the slope column, and let  $X_i = [1, t + \frac{1}{2}t^2, t^2]$ . The ratios of the three columns to  $t$  are  $1/t$ ,  $1 + t/2$  and  $t$ , none constant, so the column-by-column search returns nothing. Yet  $d = (0, 1, -\frac{1}{2})$  gives  $X_i d = t$ , so  $v = -\mathbf{1}_N$  satisfies (L) and (T), and  $\mathcal{A}_c \neq \{0\}$ .

The column space in Example 1 is  $\text{span}\{1, t, t^2\}$ , what an orthogonal-polynomial or natural-

spline mean in time produces (the monomial spelling `time + I(time^2)` is caught, because the time column appears on its own). With such a mean and a random slope, the column-by-column search silently leaves the slope degeneracy in place. By Proposition 1(a) the absorbable set depends only on the column space, and the null-space construction of Proposition 1(b), which `jmjax` now uses, does not depend on how that space is spanned.

Finally, let  $Q_c$  be an orthonormal basis of  $\mathcal{A}_c$  and  $Q \in \mathbb{R}^{N \times k}$  an orthonormal basis of  $\text{span} \bigcup_c \mathcal{A}_c$ . In the simulation design of Section 6 (an intercept, time, age and sex, with a random intercept and slope)  $\mathcal{A}_0 = \text{span}\{\mathbf{1}, \text{age}, \text{sex}\}$  and  $\mathcal{A}_1 = \text{span}\{\mathbf{1}\}$ , so  $k = 3$ .

## 4 The rotation and the metric

### 4.1 Construction and exactness

Complete  $Q$  to an orthogonal matrix  $R = [Q \mid Q_\perp] \in \mathbb{R}^{N \times N}$ . We use the Householder completion:  $R$  is stored as  $k$  reflectors, so applying  $R$  or  $R^\top$  to a vector costs  $O(Nk)$  and no  $N \times N$  matrix is formed. Instead of  $B_{\text{std}}$ , sample

$$B_{\text{std}} = R \begin{bmatrix} U \\ V \end{bmatrix} = QU + Q_\perp V, \quad B = B_{\text{std}} L^\top, \quad (5)$$

where  $Q \in \mathbb{R}^{N \times k}$ ,  $Q_\perp \in \mathbb{R}^{N \times (N-k)}$ , and  $U \in \mathbb{R}^{k \times q}$  and  $V \in \mathbb{R}^{(N-k) \times q}$  have iid  $\mathcal{N}(0, 1)$  entries.  $Q$ , and hence  $R$ , is computed once from the design matrices before sampling and is held fixed; it does not depend on any sampled parameter.

**Proposition 3** (Exactness). *For any fixed orthogonal  $R$ : (i) the map  $(U, V) \mapsto B_{\text{std}}$  is linear and volume-preserving; (ii) the prior it induces on  $B_{\text{std}}$  is the original iid  $\mathcal{N}(0, 1)$  prior; (iii) the posterior of  $(B, \theta)$  is identical to that of the non-centered parameterization (2), for every value of  $L$ ; and (iv)  $U = Q^\top B_{\text{std}}$ .*

*Proof.*  $\text{vec}(B_{\text{std}}) = (I_q \otimes R) \text{vec}([U; V])$  and  $I_q \otimes R$  is orthogonal, which gives (i), and (ii) because  $(I_q \otimes R) \mathcal{N}(0, I_{Nq}) = \mathcal{N}(0, I_{Nq})$ . The likelihood depends on  $(B_{\text{std}}, \theta)$  only, and their joint prior is unchanged, so the posterior pushed forward to  $(B, \theta)$  is unchanged: (iii). (iv) follows from  $Q^\top Q_\perp = 0$ .  $\square$

Because the target is unchanged, NUTS and its warm-up adaptation are unmodified; the rotation changes only which linear combinations of the random effects are sampling coordinates, and therefore what an adapted metric can represent.

Two choices in (5) matter. First, the rotation acts on the sampled, standardized coordinates: rotating centered effects  $R^\top B$  would reintroduce the dependence on  $\sigma_b$  that non-centering removes. Second,  $R$  acts on the subject index, the same for every column, while  $L$  acts on the column index. In vectorized form,  $\text{vec}(B) = (L \otimes I_N) \text{vec}(B_{\text{std}})$  and  $\text{vec}(B_{\text{std}}) = (I_q \otimes R) \text{vec}([U; V])$ , and

$$(L \otimes I_N)(I_q \otimes R) = L \otimes R = (I_q \otimes R)(L \otimes I_N) :$$

the two factors act on different indices and commute, which is why the construction is exact for every value of the covariance. A fixed rotation aligning particular intercept-slope combinations with coordinate axes would align them only at one value of  $L$ .

For a random intercept alone ( $q = 1$ ), `jmx` samples the centered effects  $b \sim \mathcal{N}(0, \sigma_b^2 I_N)$  directly, and the rotation is applied to  $b$ . Because the covariance is a multiple of the identity,  $R^\top b$  has the same prior for every  $\sigma_b$  and Proposition 3 holds unchanged.

## 4.2 Where the degeneracy goes

**Lemma 1** (Localization). *Let  $v \in \mathcal{A}_c$  with witness  $d$ . In the coordinates  $(\beta, U, V)$ , the likelihood-invariant shift  $(\beta, B e_c) \mapsto (\beta + d, B e_c + v)$  is*

$$\beta \mapsto \beta + d, \quad U \mapsto U + (Q^\top v) e_c^\top (L^\top)^{-1}, \quad V \mapsto V,$$

for every value of  $L$ . Every such shift therefore lies in the coordinate block  $(\beta, U)$ , of dimension  $p + kq$ .

*Proof.* Adding  $v$  to column  $c$  of  $B = B_{\text{std}} L^\top$  adds  $\Delta = v e_c^\top (L^\top)^{-1}$  to  $B_{\text{std}}$ . Every column of  $\Delta$  is a multiple of  $v \in \text{span}(Q)$ , so  $Q_\perp^\top \Delta = 0$  and  $Q^\top \Delta = (Q^\top v) e_c^\top (L^\top)^{-1}$ .  $\square$

Without the rotation, the same shift has a component on every one of the  $N$  rows of  $B_{\text{std}}$ . With it, the invariant directions mix the columns of  $U$  in proportions set by the sampled  $L$ , but stay inside the fixed block  $(\beta, U)$ ; rotating the intercept column alone would capture them only for uncorrelated random effects.  $U$  has  $kq$  coordinates, a few of which may not be invariant for their own column (in the design of Section 6, 4 of the 6 are invariant and 2 are not); including them in the dense block costs little.

### 4.3 The ridge, and the dense block

Along each invariant direction the priors still locate the split, so the posterior is proper but weakly identified. The next result describes the ridge in one plane.

**Proposition 4** (The ridge). *Consider one invariant direction in the reduced model that holds every other coordinate fixed. Let  $\tilde{\beta}$  be the coordinate of  $\beta$  along the witness, with a flat prior, and  $u$  the matching coordinate of  $U$ , with prior  $\mathcal{N}(0, 1)$ , and suppose the likelihood informs only  $s = \tilde{\beta} + \kappa u$ , with precision  $\tau$ . Then*

$$(i) \text{ corr}(\tilde{\beta}, u \mid y) = -\{1 + 1/(\tau\kappa^2)\}^{-1/2};$$

(ii) *under the ideal diagonal metric, equal to the exact marginal posterior variances (the quantity warm-up adaptation estimates), the whitened covariance of the plane has eigenvalues  $1 \pm |r|$ ,  $r$  the correlation in (i). Its wide direction has variance at most 2, but its narrow direction has variance  $1 - |r| \approx 1/(2\tau\kappa^2)$ ;*

(iii) a dense metric over  $(\tilde{\beta}, u)$  whitens the plane exactly;

(iv) for the mean direction of the random intercepts,  $\kappa = \sigma_0/\sqrt{N}$  and  $\tau = \sum_i n_i/\sigma_e^2$ , so  $\tau\kappa^2 = \bar{n}\sigma_0^2/\sigma_e^2$ , the information each subject carries about its own level. It does not depend on  $N$ .

*Proof.* The posterior precision of  $(\tilde{\beta}, u)$  is  $\tau(1, \kappa)^\top(1, \kappa) + \text{diag}(0, 1)$ , whose inverse is

$$\tau^{-1} \begin{pmatrix} 1 + \tau\kappa^2 & -\tau\kappa \\ -\tau\kappa & \tau \end{pmatrix};$$

this gives (i). Whitening a  $2 \times 2$  covariance by its diagonal gives its correlation matrix, whose eigenvalues are  $1 \pm |r|$ : (ii). (iii) is the definition of a dense metric. For (iv), moving the mean direction by  $u$  moves every subject's level by  $\sigma_0 u/\sqrt{N}$ , and the data inform a common shift of all levels with precision  $\sum_i n_i/\sigma_e^2$ .  $\square$

Heuristically, for a Gaussian target the narrow direction bounds the stable leapfrog step size and the wide one sets the trajectory length, so the number of leapfrog steps per independent draw grows like  $\{(1 + |r|)/(1 - |r|)\}^{1/2} \approx 2\sqrt{\tau\kappa^2}$  (Neal 2011). Proposition 4 is exact for the mean intercept direction, whose witness is  $\beta_0$ ; for other directions the witness mixes several coordinates of  $\beta$ , and (ii)–(iii) describe the whole  $(\beta, U)$  block. The rotation alone thus concentrates the direction in a few coordinates, but under a diagonal metric it is still narrow, most so with many visits per subject or little noise. A dense block over  $(\beta, U)$  whitens it. Without the rotation, the same fix needs a dense metric over all  $Nq$  random-effect coordinates, with  $O(N^2q^2)$  storage and work per metric-vector product; Section 7 measures this alternative.

Figure 1 shows this in the exact Gaussian posterior of the longitudinal submodel with the variance components fixed at the simulation values of Section 6. The conditional correlation of  $\beta_0$  and the mean intercept direction is  $-0.97$  unrotated and still  $-0.97$  after rotation

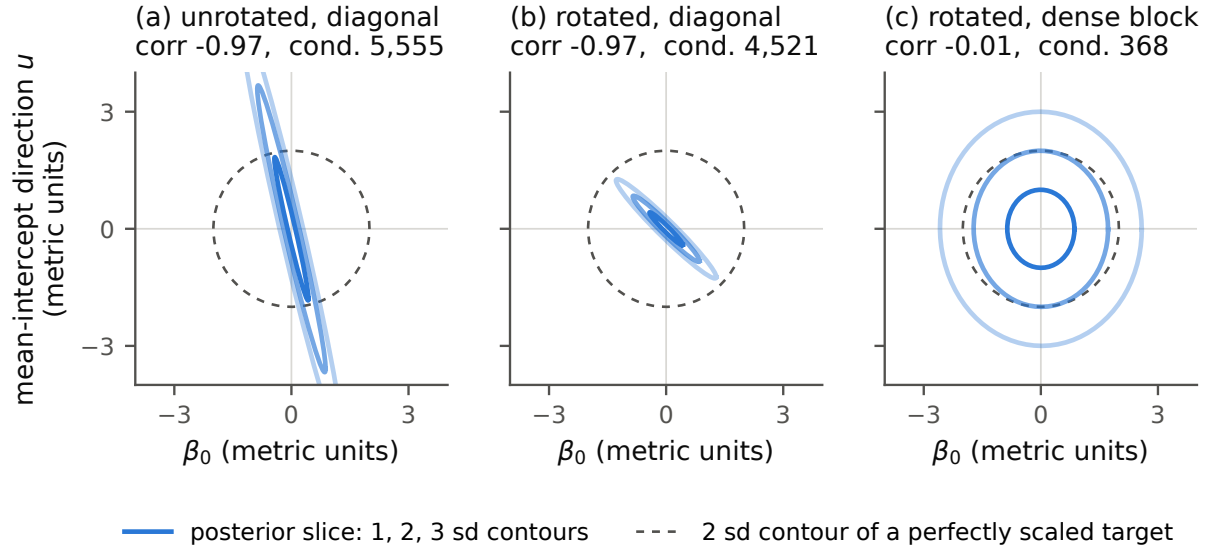


Figure 1: The location ridge and what removes it. Exact conditional posterior of  $\beta_0$  and the mean intercept direction  $u$  in the units the HMC metric sees them (Gaussian longitudinal submodel, variance components fixed;  $N = 300$ , 2–11 visits,  $\rho = 0.3$ ,  $\sigma_e = 0.3$ ). Solid: 1, 2 and 3 standard deviation contours; dashed: the 2 standard deviation contour of a perfectly scaled target. (a) Unrotated, diagonal metric. (b) Rotated, diagonal metric:  $u$  is a single coordinate but the ridge remains. (c) Rotated, dense block over  $(\beta, U)$ . “cond.”: condition number of the whole whitened posterior.

with a diagonal metric; the dense block brings it to  $-0.01$ . The condition number of the whole whitened posterior, the ratio of the extreme eigenvalues of  $M^{-1/2}\Sigma M^{-1/2}$  with  $\Sigma$  the exact posterior covariance and  $M$  the metric (diag  $\Sigma$ , or diag  $\Sigma$  with its  $(\beta, U)$  block replaced by that of  $\Sigma$ ), falls from 4,521 to 368, and by factors of 5 to 70 across the designs examined; what remains is conditioning any fit has, such as the correlation between each subject’s own intercept and slope.

## 4.4 What a fixed metric cannot do

By Lemma 1, the ridge’s orientation depends on the sampled  $L$ : on  $\sigma_0$  through  $\kappa \propto \sigma_0$  and, for directions shared between the columns, on  $\rho$ , while NUTS adapts one fixed metric. For the  $\sigma_0$  part the loss can be estimated. Let  $\delta$  be the posterior coefficient of variation of  $\sigma_0$ , roughly  $1/\sqrt{2N}$  in an idealized variance-component model with  $N$  groups (we use that value only to set the scale). A relative change  $\delta$  in  $\kappa$  tilts the plane’s narrow direction, in the units of a metric fitted at the original value, by an angle of order  $\delta\sqrt{\tau\kappa^2}$ , so a metric that whitens the plane exactly at one value of  $\sigma_0$  leaves a condition number of  $1 + O(\delta\sqrt{\tau\kappa^2})$  elsewhere. With  $\delta \approx (2N)^{-1/2}$  and  $\tau\kappa^2 = \bar{n}\sigma_0^2/\sigma_e^2$  this is

$$1 + O\left(\sqrt{\bar{n}\sigma_0^2/(\sigma_e^2 N)}\right),$$

which is small when there are many subjects relative to the information each carries, the usual regime for longitudinal data. In the exact Gaussian model of Figure 1 the worst block condition number over  $\sigma_0(1 \pm \delta)$  was 1.76, 1.50 and 1.31 for  $N = 150, 300$  and  $600$  with 2–11 visits, and 2.71, 2.05 and 1.64 with 8–20 visits. The analysis does not cover the  $\rho$  part, which grows as  $|\rho| \rightarrow 1$ ; Section 6.4 stress-tests both regimes. For  $q = 1$  in centered coordinates only the scale of the rotated coordinate’s prior involves  $\sigma_b$ , a mismatch of order  $\delta$ .

## 4.5 Implementation and cost

jmjax implements (5) in NumPyro:  $U$  and  $V$  are the sampled sites,  $B_{\text{std}}$  is computed from them with  $q$  applications of the  $k$  reflectors, and the mass matrix is dense on  $(\beta, U)$  and diagonal elsewhere, using NumPyro’s dense blocks over groups of sites, adapted during warm-up as usual. The extra cost per gradient is  $O(Nkq)$ . In the design of Section 6 the dense block has  $p + kq = 10$  coordinates, and wall time per fit was 0.83–1.10 times that of the unrotated fit. The rotation and the dense block are the package’s default for NUTS

fits with a random intercept only, or with  $q \geq 2$  correlated random effects.

## 5 Blocked samplers and the association parameter

The location degeneracy does not involve  $\alpha$ . Yet on  $\alpha$ , `jmx`'s NUTS was 3–8 times more efficient per second than `JMbayes2` even without the rotation: 5.7–8.1 times across a replicated simulation sweep with 1 to 12 covariates (each package's own ESS estimator), and 3.5 and 8.2 times on the two datasets of Section 8 (one estimator for both). `JMbayes2` updates the parameters in blocks, with adaptive random-walk Metropolis steps for the coefficients. We argue that, on the two datasets measured, about three quarters (on a log scale) of the ESS-per-draw gap on  $\alpha$  is what even an ideal two-block sampler would incur.

Write  $\psi$  for every parameter other than  $\alpha$ : the random effects,  $\beta$ ,  $\gamma$ , the baseline-hazard coefficients and the variance components.

**Proposition 5** (Blocked updates of  $\alpha$ ). *Consider the two-block sampler that draws  $\alpha$  exactly from  $p(\alpha \mid \psi, y)$  and then  $\psi$  exactly from  $p(\psi \mid \alpha, y)$ , and let*

$$r = \frac{\text{Var}\{E(\alpha \mid \psi, y)\}}{\text{Var}(\alpha \mid y)} = 1 - \frac{E\{\text{Var}(\alpha \mid \psi, y)\}}{\text{Var}(\alpha \mid y)},$$

*the fraction of  $\alpha$ 's posterior variance explained by the other parameters, and  $R^2 \leq r$  the squared multiple correlation of  $\alpha$  on  $\psi$ .*

- (i) *At stationarity the lag-1 autocorrelation of the  $\alpha$  chain is  $r$ .*
- (ii) *Its integrated autocorrelation time is at least  $(1+r)/(1-r)$ , so its effective sample size per draw is at most  $(1-r)/(1+r) \leq (1-R^2)/(1+R^2)$ .*
- (iii) *If the posterior is Gaussian,  $r = R^2$ , the  $\alpha$  chain is  $AR(1)$  with coefficient  $R^2$ , and the bound in (ii) is attained.*

*Proof.* The  $\alpha$  chain of a two-block sampler is a data-augmentation chain (Liu et al. 1994).

Let  $P$  map a function  $h$  of  $\alpha$  to the function  $(Ph)(\psi) = \mathbb{E}\{h(\alpha) \mid \psi, y\}$ , and  $P^*$  map  $g$  to  $(P^*g)(\alpha) = \mathbb{E}\{g(\psi) \mid \alpha, y\}$ ; these are adjoint in the  $L^2$  inner products of the posterior. The transition operator of the  $\alpha$  chain is  $K = P^*P$ , so  $K$  is self-adjoint and  $\langle Kh, h \rangle = \|Ph\|^2 \geq 0$ . With  $h(\alpha) = \alpha - \mathbb{E}(\alpha \mid y)$ , the lag-1 autocovariance is  $\langle Kh, h \rangle = \text{Var}\{\mathbb{E}(\alpha \mid \psi, y)\}$ , which gives (i); the second expression for  $r$  is the law of total variance, and  $R^2 \leq r$  because the best linear predictor of  $\alpha$  explains no more variance than  $\mathbb{E}(\alpha \mid \psi, y)$ . For (ii), the spectral theorem writes the lag- $k$  autocorrelation of  $\alpha$  as  $\rho_k = \int \lambda^k \mu(d\lambda)$  for a probability measure  $\mu$  on  $[0, 1]$ , so Jensen's inequality gives  $\rho_k \geq \rho_1^k = r^k$  and  $1 + 2 \sum_{k \geq 1} \rho_k \geq (1+r)/(1-r)$ ; the function  $(1-x)/(1+x)$  is decreasing. For (iii),  $\mathbb{E}(\alpha \mid \psi)$  is linear in  $\psi$  and  $\mathbb{E}(\psi \mid \alpha)$  linear in  $\alpha$ ; composing the two regressions gives  $\alpha_{t+1} = c + R^2 \alpha_t + \varepsilon_t$  with  $\varepsilon_t$  independent of the past, since the coefficient is  $\text{Cov}(\alpha, \psi) \Sigma_\psi^{-1} \text{Cov}(\psi, \alpha) / \text{Var}(\alpha) = R^2$ , and then  $\rho_k = (R^2)^k$ .  $\square$

Part (ii) makes a measured linear  $R^2$  usable: it bounds the efficiency of any exact two-block sampler from above, whatever the shape of the posterior. The proposition applies known theory; the two-block sampler is a benchmark, not a model of JMBayes2, and finer blocking and random-walk Metropolis steps can be expected to make a real implementation slower still (Liu et al. 1994, Roberts & Sahu 1997), though not in every case. The contrast with HMC, which moves  $\alpha$  and  $\psi$  together, is a heuristic. Under a fixed metric a correlation between them is paid for in leapfrog steps, which for a Gaussian target grow with the square root of the condition number (Neal 2011). A single correlation at level  $R^2$  contributes a condition number  $(1 + \sqrt{R^2})/(1 - \sqrt{R^2})$ , so the cost grows at most like  $1/\sqrt{1 - R^2}$ , against  $1/(1 - R^2)$  for the blocked sampler, and often not at all, because in a joint model the step size is already set by some other narrow direction. Section S1 of the Supplementary Material checks both statements in a Gaussian model.

## 5.1 How large $R^2$ is in a joint model

Two mechanisms can make  $R^2$  large. The first is errors in variables: given the random effects,  $\alpha$  is the coefficient of a known time-varying covariate, while marginally it also carries the uncertainty in every trajectory  $m_i(t)$ . The second is the level of the hazard: a common shift of every  $m_i(t)$  changes  $\alpha m_i(t)$  by an amount the log baseline hazard absorbs, so  $\alpha$  is correlated with the baseline-hazard level unless the marker is centered.

We estimated the linear explained variance  $R^2$  by an ordinary least-squares regression of the 20,000 draws of  $\alpha$  from a long rotated fit (4 chains of 5000 draws, close to independent) on every other sampled quantity (960 regressors for `aids` and 649 for `pbc2`), and report the adjusted  $R^2$ , which differs from the unadjusted value by less than 0.01. We also regressed  $\alpha$  on each block alone. On `aids`,  $R^2 = 0.873$ ; the survival block alone ( $\gamma$  and the baseline-hazard coefficients) explained 0.86 of  $\alpha$ 's posterior variance and the random effects alone 0.30. On `pbc2` the figures were 0.853, 0.82 and 0.20: in these two datasets, most of the explained variance is attributable to the survival block rather than the random effects. By Proposition 5,  $R^2 \leq r$ , so the efficiency it implies for an ideal blocked sampler is an upper bound.

Figure 2 puts the measurements against the prediction. At the measured  $R^2$  an ideal two-block sampler keeps at most 0.068 (`aids`) and 0.079 (`pbc2`) effective draws per draw, one in 13–15. JMBayes2 achieved 0.032 and 0.025 (its draws from the runs of Section 8, NumPyro's estimator; its own gives 0.035 and 0.028), and `jmxax`'s NUTS 1.05 and 1.72 in the long fit. On a log scale, the ideal benchmark accounts for 78% of the measured ESS-per-draw gap on `aids` (15.4 of 33 times) and 73% on `pbc2` (21.7 of 68 times); the rest, a factor of two to three, is attributable to the implementation. A blocked sampler could escape most of the structural part by updating  $\alpha$  jointly with  $\gamma$  and the baseline-hazard coefficients: with only the random effects conditioned away, the ideal ESS per draw would

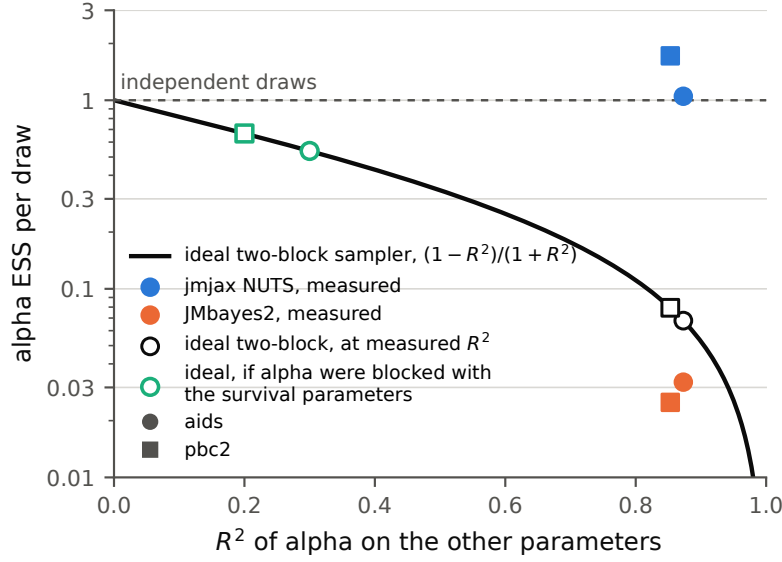


Figure 2: Effective sample size per draw for  $\alpha$ . Curve: an ideal two-block sampler,  $(1 - R^2)/(1 + R^2)$  (Proposition 5). Black open markers: the ideal at the measured  $R^2$ ; green open markers: the ideal if  $\alpha$  were updated in one block with the survival parameters ( $R^2 = 0.30$  and  $0.20$ ).

be about 0.54 and 0.67. But that block has 10–20 strongly correlated dimensions without exact conditional draws, and an adaptive random-walk proposal on it is itself slow. HMC moves  $\alpha$  and  $\psi$  jointly without a designed block, although how well it does so still depends on the metric and on the rest of the geometry.

## 6 Simulation

### 6.1 Design

Data were simulated from a joint model with a random intercept and slope, longitudinal mean  $\beta_0 + \beta_{\text{time}}t + \beta_{\text{age}}\text{age}_i + \beta_{\text{sex}}\text{sex}_i$  with  $\beta_0 = 2$  and  $\beta_{\text{time}} = 0.5$ ,  $\text{age}_i \sim \mathcal{N}(0, 10^2)$  and  $\text{sex}_i \sim \text{Bernoulli}(0.5)$ ,  $(\sigma_0, \sigma_1) = (0.8, 0.3)$ ,  $\sigma_e = 0.3$ , and a Weibull hazard (shape 1.3) with  $\alpha = 0.5$  and a treatment effect  $\gamma = -0.3$ , calibrated to an event rate of about one

half. Visits were scheduled at regular intervals up to  $t = 10$  and stopped at the event or censoring time. All fits used the spline baseline hazard of Section 2.1, so the fitted model does not contain the true hazard exactly. Each design was simulated with three seeds, each fitted unrotated and with the rotation and its dense block. ESS is NumPyro’s (version 0.15) multi-chain autocorrelation estimator, truncated by Geyer’s initial monotone sequence and without rank normalization (Geyer 1992, Gelman et al. 2013), and  $\hat{R}$  is its split-chain version. ESS per second divides by the wall time of the whole run, warm-up and compilation included; ESS per gradient divides by the number of leapfrog steps in the sampling phase. Ratios are rotated over unrotated, paired by seed, with geometric means over seeds.

The core grid crossed about 9 against about 17 visits per subject with  $\rho \in \{0, 0.3, 0.6, 0.9\}$  at  $N = 300$ , and added  $N \in \{150, 600\}$  at 9 visits and  $\rho = 0.3$ : ten designs, 2 chains of 500 warm-up and 1000 sampling iterations.

## 6.2 Efficiency

Over all 30 paired fits of Table 1, the geometric-mean gain in ESS per second was 13.5 for the intercept (per seed 4.5–31.6) and 21.1 for the time slope (5.1–70.8). Wall time was essentially unchanged (ratio 1.03, range 0.83–1.10) and the rotated fit took 0.89 times as many leapfrog steps per iteration, so per gradient the gains were 15.6 and 24.3. Almost all of the gain is in ESS per draw: a median of about 1.7 rotated against 0.1 unrotated. There were no divergences, and the worst  $\hat{R}$  on a reported coefficient was 1.002 rotated and 1.044 unrotated. Posterior means agreed to a median of 0.027 posterior standard deviations (largest paired difference 2.7 Monte Carlo standard errors over 90 pairs), and posterior standard deviations to within Monte Carlo error (ratios 1.01 for the intercept and 0.98 for the time slope).

Table 1: Core grid: ESS per second for the intercept and the time slope (means over 3 seeds), and the rotated/unrotated ratio (geometric mean of the paired per-seed ratios).

| visits | $\rho$ | $N$ | intercept |         |       | time slope |         |       | $\alpha$ |
|--------|--------|-----|-----------|---------|-------|------------|---------|-------|----------|
|        |        |     | unrot.    | rotated | ratio | unrot.     | rotated | ratio | ratio    |
| 9      | 0.0    | 300 | 25.9      | 354.4   | 14.2  | 9.1        | 304.9   | 36.4  | 0.85     |
| 17     | 0.0    | 300 | 12.1      | 229.4   | 19.3  | 4.0        | 234.9   | 59.2  | 1.32     |
| 9      | 0.3    | 300 | 30.0      | 253.3   | 8.6   | 19.0       | 251.1   | 13.4  | 0.72     |
| 17     | 0.3    | 300 | 11.3      | 248.5   | 22.2  | 6.4        | 262.5   | 41.9  | 1.33     |
| 9      | 0.6    | 300 | 26.1      | 245.7   | 9.4   | 29.9       | 229.3   | 7.7   | 1.00     |
| 17     | 0.6    | 300 | 9.6       | 231.6   | 24.0  | 10.0       | 251.2   | 24.9  | 1.59     |
| 9      | 0.9    | 300 | 20.4      | 220.7   | 11.1  | 20.3       | 250.3   | 12.8  | 0.90     |
| 17     | 0.9    | 300 | 8.9       | 218.9   | 24.9  | 8.6        | 217.7   | 25.6  | 1.59     |
| 9      | 0.3    | 150 | 41.8      | 323.7   | 7.6   | 32.6       | 420.4   | 13.1  | 0.76     |
| 9      | 0.3    | 600 | 15.9      | 132.0   | 8.2   | 8.8        | 151.7   | 17.7  | 0.77     |

The pattern follows Proposition 4. The gain is largest with dense visits, where each subject carries the most information about its own random effects (19–25 times for the intercept, 25–59 for the slope), and does not grow with  $N$  (7.6 at  $N = 150$ , 8.2 at  $N = 600$ ). Figure 3 shows all three grids against the per-subject information  $\bar{n} \sigma_0^2 / \sigma_e^2$ . The association parameter, which the degeneracy does not involve, split by design: 0.72–1.00 in the six designs with about 9 visits and 1.32–1.59 in the four with about 17 (geometric mean 1.04). We have no established explanation; on real data (Section 8) the effect on  $\alpha$  tracked whether the degeneracy had been setting NUTS’s trajectory length, a plausible reading here too that was not measured.

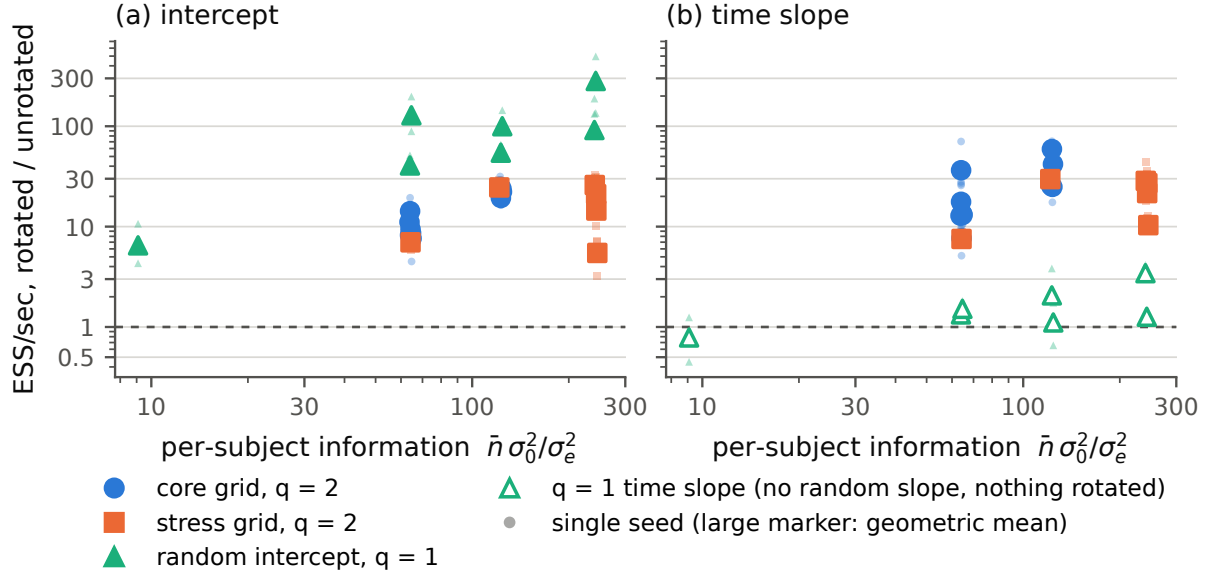


Figure 3: ESS per second of the reported coefficient, rotated over unrotated fit on the same simulated data, against the per-subject information  $\bar{n} \sigma_0^2 / \sigma_e^2$  ( $\sigma_0 = 0.8$ ). Large markers: geometric mean over three seeds in one design; small: single seeds. Core and stress grids have a random intercept and slope ( $q = 2$ ); the  $q = 1$  grid has a random intercept only. In (b), open triangles are the  $q = 1$  time slope, which has no random effect and so nothing to rotate. The  $q = 1$  intercept ratios are at the standard run length, where most unrotated fits had not converged; Section 6.3 compares converged fits.

### 6.3 Random intercept only

The same simulator with  $\sigma_1 = 0$ , fitted with a random intercept only, gave seven designs (about 9, 17 and 35 visits,  $N \in \{100, 300\}$ , and one with  $\sigma_e = 0.8$ ). The intercept gained 6.5–282 times and the coefficient of age 6.6–207 times, but at this run length 17 of the 21 unrotated fits had some  $\hat{R}$  above 1.05, while every rotated fit had  $\hat{R} \leq 1.004$ . For a comparison between converged fits we reran three designs ( $N = 300$ ; about 9, 17 and 35 visits), extending the unrotated fit to up to 32 times the standard length, with a criterion fixed in advance:  $\hat{R} \leq 1.01$  on every monitored parameter and  $\text{ESS} \geq 400$  for the intercept,

age, the time slope and  $\alpha$ . Every rotated fit met it at the standard length. The unrotated fit met it first at 16 times the standard length with 9 visits, at 16 and 32 times with 17 visits, and not by 32 times with 35 visits. Its ESS per draw for the intercept did not change with run length, and the rotated fit's was of order 100 times larger, so the advantage is a property of the stationary chains, not of short runs. In wall time to the criterion the unrotated fit needed 3.5 and 6.5–6.6 times as long in the first two designs, and more than 10–14 times in the third (Section S2 of the Supplementary Material).

## 6.4 Stress grid

Section 4.4 names two regimes where the rotation should be weakest: few subjects each carrying a lot of information, and strongly correlated random effects. A stress grid of six designs targeted both (about 35 visits with  $N \in \{50, 100\}$  and  $\rho \in \{0.3, 0.95\}$ , plus two designs with  $\rho = 0.95$ ), with the criterion, fixed in advance, that the rotation must not fall below the unrotated fit on the intercept or time slope in any design. It did not: the lowest design gained 5.5 times and the lowest single seed 3.2 times, and per gradient the geometric-mean gains were 16.7 for the intercept and 22.0 for the time slope (Section S3 of the Supplementary Material). The design with the fewest subjects and the most information per subject gained least, as the fixed-metric residual predicts, and strong correlation did not reduce the gains. The predicted weakness is visible only against a variant that removes the absorbable directions and restores the reported coefficients with an exact correction (Section 9.1), which has no fixed-metric residual: in the 35-visit designs it was 1.4–2 times more efficient than the rotation, and about equal with 9 visits.

## 6.5 Calibration

Proposition 3 says the rotation cannot change the posterior, so a calibration study checks the implementation. With  $N = 300$  and 100 replicates, none excluded, coverage of 95% intervals agreed between the two fits within 0.01 and the ratio of mean posterior standard deviation to the standard deviation of posterior means within 0.015 for  $\beta_0$ ,  $\beta_{\text{time}}$ ,  $\alpha$  and  $\sigma_0$  (Section S4 of the Supplementary Material). Both fits showed an upward bias of about 3 Monte Carlo standard errors (2.8%) in  $\alpha$ , which fell to 1.1% at  $N = 1200$ , consistent with ordinary finite-sample bias.

## 7 Beyond joint models

Nothing in Sections 3–4 depends on the survival submodel except Condition S. To check that the gains are not specific to joint models, we applied the construction, in a standalone NumPyro program, to two classical mixed models. For `Orthodont` (Potthoff & Roy 1964, Pinheiro & Bates 2000), 27 children measured at 4 ages, a Gaussian linear mixed model (LMM)  $\text{distance} \sim \text{age} \times \text{male} + (1 + \text{age} \mid \text{child})$  with age centered has absorbable directions  $\{\mathbf{1}, \text{male}\}$  in both random-effect columns ( $k = 2$ ). For the toenail onychomycosis trial (De Backer et al. 1998, Lesaffre & Spiessens 2001), 1908 binary outcomes on 294 patients, a logistic generalized linear mixed model (GLMM)  $y \sim \text{trt} \times \text{time} + (1 \mid \text{patient})$  has absorbable directions  $\{\mathbf{1}, \text{trt}\}$ . Both use the non-centered parameterization, as brms and rstanarm do by default, with  $\mathcal{N}(0, 10^2)$  priors on  $\beta$ , half-normal(5) priors on the standard deviations and an LKJ(2) prior on the correlation. Five arms were run on each (4 chains of 1000 warm-up and 1000 sampling iterations, 3 seeds): *unrotated* with a diagonal metric (the reference); *rotated*, *diagonal*, the rotation alone; *rotated*, *dense block*, the method; *unrotated*, *full dense*, a dense metric over every sampled coordinate, the obvious alternative way to learn the ridge; and *centered*, sampling  $b_i \sim \mathcal{N}(0, LL^\top)$  with a diagonal metric, the

Table 2: Two mixed models: ESS per 1000 gradient evaluations relative to the unrotated fit with a diagonal metric, and in brackets ESS per second (geometric means over 3 seeds; ranges over the coefficients in a group). The last three columns give leapfrog steps per draw, seconds per run (whole run) and divergences over the 3 seeds.

| arm                                                                                                           | coefficients              |                   |                   | steps | sec   | div. |
|---------------------------------------------------------------------------------------------------------------|---------------------------|-------------------|-------------------|-------|-------|------|
|                                                                                                               | intercept, subject-const. | absorbed by slope | variance comp.    |       |       |      |
| Orthodont (LMM, $q = 2$ , $N = 27$ ; subject-constant: male; slope-absorbed: age, age $\times$ male)          |                           |                   |                   |       |       |      |
| unrotated                                                                                                     | 1                         | 1                 | 1                 | 30.7  | 1.9   | 0    |
| rotated, diagonal                                                                                             | 2.3–2.4 (2.6–2.7)         | 0.9 (1.1)         | 0.9 (1.0–1.1)     | 30.9  | 1.6   | 0    |
| rotated, dense block                                                                                          | 7.1 (4.7)                 | 2.4–2.5 (1.6)     | 1.3–1.6 (0.8–1.0) | 16.5  | 1.5   | 0    |
| unrotated, full dense                                                                                         | 5.5–5.6 (2.0–2.1)         | 1.7–1.8 (0.6)     | 1.2–2.0 (0.5–0.7) | 18.9  | 3.0   | 2    |
| centered                                                                                                      | 0.4 (0.6–0.7)             | 0.4–0.5 (0.7)     | 0.1–1.7 (0.1–2.6) | 52.5  | 1.7   | 34   |
| toenail (logistic GLMM, $q = 1$ , $N = 294$ ; subject-constant: trt; not absorbable: time, trt $\times$ time) |                           |                   |                   |       |       |      |
| unrotated                                                                                                     | 1                         | —                 | 1                 | 31.0  | 7.7   | 0    |
| rotated, diagonal                                                                                             | 2.5–2.8 (2.3–2.5)         | —                 | 0.9 (0.9)         | 31.0  | 8.5   | 0    |
| rotated, dense block                                                                                          | 5.5–6.7 (4.2–5.1)         | —                 | 1.1 (0.9)         | 21.2  | 6.8   | 0    |
| unrotated, full dense                                                                                         | 3.2–3.7 (0.2–0.3)         | —                 | 0.9 (0.07)        | 47.6  | 166.8 | 0    |
| centered                                                                                                      | 0.8–1.1 (0.5–0.8)         | —                 | 0.4 (0.2)         | 30.6  | 10.8  | 0    |

classical remedy (Gelfand et al. 1995). Because each run takes seconds, we report ESS per 1000 gradient evaluations, which does not depend on the machine, and ESS per second of the whole run as the practical measure.

With the method, the intercept and the subject-constant coefficients gained 5.5–7.1 times per gradient and 4.2–5.1 times per second, and the coefficients absorbed by the random slope in **Orthodont** 2.4–2.5 times per gradient (Table 2). The rotation alone supplied 2.3–2.8 on the intercept and subject-constant coefficients and nothing on the slope-absorbed ones; the dense block supplied the rest, as Proposition 4 predicts. The coefficients with no

absorbable direction (toenail’s time and treatment-by-time), a negative control, moved by 1.3–1.4 per gradient and 1.0 per second.

**Why not a full dense metric?** In `Orthodont`, with  $N = 27$  and 62 sampled coordinates, a full dense metric recovered most of the gain per gradient (5.5–5.6 against 7.1 for the intercept and male) and did somewhat better on the variance components, but at twice the cost per run, so per second it gained 2.0–2.1 against the method’s 4.7. In the toenail model, with  $N = 294$  and about 300 coordinates, it was worse than the method on every measure: a 300-dimensional metric estimated from 1000 warm-up iterations needed 47.6 leapfrog steps per draw against 21.2, per gradient it gained 3.2–3.7 against 5.5–6.7, and each run took 167 seconds against 7, so per second it was 0.2–0.3 times the *unrotated* fit. This is a result for the tested adaptation budget, not a limit of dense NUTS: a longer warm-up would estimate the metric better, at a cost that grows with the number of coordinates. The rotation makes a dense metric affordable by moving the ridge into  $p + kq$  coordinates (8 and 6 here), whatever  $N$  is.

**Why not center?** In these two configurations centering did not remove the degeneracy’s cost (per gradient 0.4 on the `Orthodont` intercept and male coefficients, 0.8–1.1 on toenail’s) and brought back the funnel: in `Orthodont` it produced 34 divergent transitions over the 3 seeds,  $\hat{R}$  up to 1.047 and 52.5 leapfrog steps per draw, and the slope standard deviation mixed at 0.06 times the unrotated rate; in toenail the random-effect standard deviation mixed at 0.35 times. Both models carry little information per subject, the regime in which centering is known to do poorly ([Gelfand et al. 1995](#), [Browne et al. 2009](#)); where subjects are informative, centering can work well, as JMBayes2’s centered coefficients do in Section 8. The rotation needs no choice between the two and, in every design tested, reduced the location cost within the non-centered parameterization.

Across all arms, posterior means agreed with the unrotated fit's to within 0.14 posterior standard deviations. Apart from the centered arm,  $\hat{R} \leq 1.011$  and there were at most 2 divergences (full dense, `Orthodont`).

## 8 Real data

### 8.1 Rotated against unrotated

We fitted two standard joint-model datasets, both distributed with JMBayes2, unrotated and rotated from the same preliminary mixed-model and Cox fits and sampler seed, with 4 chains of 1000 warm-up and 1000 sampling iterations and 3 seeds per dataset. For `aids` (Abrams et al. 1994, Guo & Carlin 2004), 467 patients, the marker model is  $\sqrt{\text{CD4}} \sim \text{obstime} + \text{drug} + \text{gender}$ , with three absorbable intercept directions (constant, drug, gender) and one slope direction. For `pbc2` (Murtaugh et al. 1994), 312 patients, it is  $\log(\text{serum bilirubin}) \sim \text{year} + \text{age}$ , with two absorbable intercept directions (constant, age) and one slope direction. Both have a random intercept and slope on time and a hazard for death depending on drug.

The intercept and subject-constant coefficients gained 17–28 times, and the time slope, also on an absorbable direction, 2.3–4.4 times (Table 3). Every posterior mean agreed between the two fits within Monte Carlo error (largest paired difference 1.65 Monte Carlo standard errors over 19 parameter-dataset pairs). The two datasets differ in one instructive way. On `aids` the unrotated trajectories were three times longer: the degeneracy was what set NUTS's trajectory length, and removing it made every iteration cheaper, so every parameter gained,  $\alpha$  included, and sampling time fell by 40%. On `pbc2` the trajectory length was set by something else, and the parameters the degeneracy does not involve changed by 0.82–1.28 times ( $\alpha$  0.95).

Table 3: Real data: ESS per second, rotated over unrotated, mean over 3 seeds (per-seed range in brackets), and sampler statistics.

|                                        | aids               | pbc2             |
|----------------------------------------|--------------------|------------------|
| intercept                              | 26.9 [21–34]       | 17.2 [12–25]     |
| subject-constant covariates            | 24.8, 28.5 [21–33] | 16.6 [12–23]     |
| time slope                             | 2.3 [2.0–2.7]      | 4.4 [4.0–4.6]    |
| $\alpha$                               | 2.0 [1.8–2.4]      | 0.95 [0.88–1.01] |
| $\sigma_e, \sigma_b, \rho, \gamma$     | 1.4–2.4            | 0.82–1.28        |
| sampling time (s), rotated / unrotated | 10.8 / 17.6        | 12.4 / 11.7      |
| leapfrog steps per iteration           | 31 / 92–111        | 60–63 / 63       |
| maximum $\hat{R}$ , all parameters     | 1.010 / 1.010      | 1.007 / 1.018    |

With a random intercept only, the same datasets show the degeneracy more starkly: the rotation raised the affected coefficients’ ESS per second 51–53 times on `aids` and 19–20 times on `pbc2` (Section S5 of the Supplementary Material).

## 8.2 A blocked sampler

JMbayes2 (Rizopoulos et al. 2026) fits the same models by Metropolis-within-Gibbs and hierarchically centers the coefficients it can, the classical remedy for the location degeneracy (Gelfand et al. 1995). It therefore offers two comparisons: on the centered coefficients, of the rotation with centering; on the rest, and on  $\alpha$  in particular, of HMC with a blocked sampler (Section 5). JMbayes2 was run with 4 chains of 16,000 iterations and 4,000 burn-in, well above its defaults, timing the whole model-fitting call. We saved the draws of all three fits (a separate run from Table 3, same settings) and computed every ESS in Table 4 with NumPyro’s estimator. With the rank-normalized bulk ESS of the `posterior` package the ratios were within 5% of those in the table, and with each package’s own estimator

Table 4: ESS per second relative to JMBayes2 (same data, geometric means over 3 seeds; one ESS estimator, NumPyro’s, for all fits).

|                               | aids      |           | pbc2      |         |
|-------------------------------|-----------|-----------|-----------|---------|
|                               | unrotated | rotated   | unrotated | rotated |
| intercept                     | 0.04      | 0.97      | 0.05      | 0.93    |
| subject-constant covariates   | 0.03–0.04 | 0.88–0.98 | 0.06      | 0.98    |
| time slope                    | 3.3       | 7.7       | 0.42      | 1.9     |
| $\alpha$                      | 3.5       | 7.2       | 8.2       | 7.9     |
| survival coefficient $\gamma$ | 1.2       | 2.8       | 2.1       | 1.8     |

within 18% (JMBayes2’s own estimator is somewhat more generous to it).

The unrotated NUTS fit trailed JMBayes2 by a factor of 17–33 on the coefficients JMBayes2 centers (Table 4). Rotated, it is at parity on them (0.88–0.98) and ahead on the others: 1.9–7.7 times on the time slope, 1.8–2.8 on  $\gamma$  and 7.2–7.9 on  $\alpha$ . The rotated NUTS fit therefore achieves efficiency comparable to JMBayes2 on the regression coefficients that JMBayes2 centers, while retaining the non-centered parameterization for the random-effect variances. The lead on  $\alpha$  is present without the rotation, and Section 5 attributes most of its per-draw part to blocked updating; the rotation added to it on **aids** (3.5 to 7.2) by shortening NUTS’s trajectories. JMBayes2 produces 4.6–7.3 times as many draws per second, but only one effective draw in 31–40 for  $\alpha$  (Figure 2).

## 9 Discussion

### 9.1 Related work

Approaches to a location degeneracy remove the directions (constraints, projections, integration), reparameterize them (centering, non-centering, interweaving), or adapt the sampler’s metric to them. The rotation is a reparameterization chosen so that the third becomes cheap, and unlike removal it keeps the original parameter space and posterior and returns the original parameters.

**Choosing a parameterization.** Centering ([Gelfand et al. 1995, 1996](#)) removes the location degeneracy and non-centering the funnel ([Papaspiliopoulos et al. 2003, 2007](#)); partial non-centering interpolates, the ancillarity-sufficiency interweaving strategy ([Yu & Meng 2011](#)) interweaves the two within a Gibbs sampler, and [Gorinova et al. \(2020\)](#) automate the choice per variable. All of these choose between centered and non-centered coordinates for each random effect. The rotation keeps the non-centered parameterization, with its benefits for the variances, and changes only which linear combinations of the standardized effects are sampling coordinates; against JMBayes2, which centers the coefficients it can, it achieved comparable efficiency on those coefficients (Section 8). [Browne et al. \(2009\)](#) studied reparameterizations for multilevel survival models under Gibbs sampling. Their orthogonal parameterization orthogonalizes the fixed-effect predictors among themselves, as Stan’s QR reparameterization does ([Stan Development Team 2026](#)), but they observe that predictors correlated with the cluster indicator vectors make fixed and random effects “play similar roles” and suggest orthogonality to those indicators as future work. The absorbable subspace is that span, handled by rotating the random effects rather than the predictors.

**Removing the directions.** [Zanella & Roberts \(2021\)](#) show that sum-to-zero constraints always improve the Gibbs sampler for crossed random effects, and that they also help NUTS dramatically on a Poisson crossed model (minimum ESS from 3.7 to 4977). Their construction samples the posterior conditioned on the constraint, and Stan’s `sum_to_zero_vector` likewise changes the prior unless compensated. Restricted spatial regression ([Reich et al. 2006](#), [Hughes & Haran 2013](#)) projects spatial random effects off the fixed-effect design, and is known to affect inference on the fixed effects ([Hodges & Reich 2010](#), [Hanks et al. 2015](#), [Khan & Calder 2022](#)). An earlier version of `jmjjax` swept the absorbable directions out of the sampled random-effect coordinates: without a correction this miscalibrated the intervals for the swept coefficients, and with an exact correction it was about as efficient as the rotation in ordinary designs and up to twice as efficient in the most information-dense ones (Section 6.4), at the price of the correction step. The closest precedent is [Vines et al. \(1996\)](#), whose transformation is exact: for independent Gaussian random effects and a Gaussian prior on the grand mean, the directions that shift the grand mean against the mean of each factor appear in no likelihood term and can be integrated out. Their method covers the grand-mean direction only, reports sum-to-zero contrasts rather than the original parameters, and leaves correlated random effects to further work; they also warn that recentering the effects inside each iteration has an unknown effect on the stationary distribution. The rotation extends the exact approach to every absorbable direction, including covariate-weighted and random-slope ones, to correlated random effects and to any prior on the fixed effects.

**Analyzing the directions.** In the discussion of [Zanella & Roberts \(2021\)](#), [Yang & Liu \(2021\)](#) decompose the Gibbs sampler for a random-intercept model with covariates using the subject-level covariate means, which span the covariate-weighted absorbable subspace for the intercept column, and show that the convergence rate is governed by the correlation

between  $\beta$  and the random effects along it; they note that the decomposition breaks down for general designs, and do not treat random slopes. In the same discussion Robert (2021) asked whether these parameterization choices had been studied for HMC, and the rejoinder (Zanella & Roberts 2021) called that open. Lemma 1 and Proposition 4 are an analysis on the HMC side for the absorbable directions. They complement general results on linear preconditioning for Markov chain Monte Carlo (Hird & Livingstone 2025): the block the rotation creates is small enough to precondition densely because the model says where the ridge is.

**Joint-model software.** JMbays2 (Rizopoulos et al. 2026) uses Metropolis-within-Gibbs; `stan_jm` in `rstanarm` (Brilleman et al. 2018) uses NUTS with non-centered random effects and so, we expect, meets the same ridge, which we have not tested; INLAjoint (Rustand et al. 2024) uses integrated nested Laplace approximations. We have not found a previous treatment that applies the data-augmentation autocorrelation identity (Liu et al. 1994) to the association parameter together with a measurement of how its shared variance divides between the random effects and the baseline hazard.

## 9.2 Limits and extensions

With about 9 visits per subject the rotation gave  $\alpha$  0.72–1.00 of the unrotated fit’s ESS per second, though it gained with denser follow-up; the mechanism is not established. In the most information-dense designs the swept variant was up to twice as efficient, as the fixed-metric residual of Section 4.4 predicts, although the rotation was still 5–30 times ahead of the unrotated fit there. The simulation evidence comes from one design family with a spline baseline hazard. The basis construction’s rank decision is a tolerance judgment when absorbable and identified directions are nearly collinear, although every returned direction is verified before use, and Condition S can fail for very flexible time bases with

few observations per subject.

A metric that follows the sampled  $\sigma_0$ , for example by rescaling the dense block with the current  $L$ , is the obvious way to remove the residual and has not been tried. The construction needs only the fixed- and random-effects designs, so it applies to any model whose random effects enter through a linear predictor, including the GLMMs of `brms` and `rstanarm`. Crossed random effects need one basis per grouping factor, each rotating that factor’s own index; the argument of Proposition 3 carries over unchanged.

### 9.3 Conclusion

The location degeneracy between fixed and random effects is shared by every regression with a random intercept and subject-constant fixed-effect columns, and it recurs for the time slope in every model with a random slope; this much is algebra. In the models studied here it slowed HMC substantially on the coefficients an analyst reports. Rotating the standardized random effects so that the absorbable directions become a small block of coordinates, and giving that block and the fixed effects one dense metric block, largely whitens the resulting ridge without changing the model. The association parameter of a joint model is a separate matter: in the two joint models measured, its posterior variance was largely shared with the survival parameters, a sharing that a sampler updating it conditionally pays for in autocorrelation, whereas HMC, heuristically, pays for it, if at all, in integration steps.

## Disclosure statement

The author reports there are no competing interests to declare.

## Declaration of generative AI use

Generative AI tools were used in this work. Anthropic’s Claude was used to help write and check analysis code, to search and summarize the literature, and to draft and revise the manuscript under the author’s direction; OpenAI’s ChatGPT was used to review drafts; Google’s Gemini was used to summarize one reference. The author designed the study, ran all computations, checked the results and claims against the code and its output, and takes full responsibility for the content.

## Data availability statement

The `aids` and `pbc2` data are distributed with the R package `JMbayes2` (Rizopoulos et al. 2026), the `Orthodont` data with `nlme` (Pinheiro & Bates 2000), and the `toenail` data with `HSAUR3` (Hothorn & Everitt 2014). The simulated data are generated by the scripts listed in Section S6 of the Supplementary Material.

## SUPPLEMENTARY MATERIAL

**Supplement:** Details of the simulation studies (runs to convergence, stress grid, calibration), a Gaussian check of the blocked-sampler comparison, the random-intercept-only fits of the real data, and a table mapping each result to the script that reproduces it (PDF).

**Code archive:** Source of the R package `jmjax` at release 0.3.0 (Guo 2026), which implements the method, with the scripts and result files that reproduce every table and figure and a README mapping each result to its script (zip file). The package is also available at <https://github.com/jmjaxmaintainer-jpg/jmjax>.

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## A Proofs

*Proof of Proposition 2.* ( $\supseteq$ ) Let  $d \in \ker M_c$  and  $v = v(d)$  as in Condition S. For  $i \in S_c$ , Proposition 1(b) gives  $X_i d + v_i z_i = 0$ ; for  $i \notin S_c$ ,  $d \in \ker M_c$  gives  $X_i d = 0$  and  $z_i = 0$ , so (L) holds. Condition S gives (T). Hence  $v \in \mathcal{A}_c$ .

( $\subseteq$ ) Let  $v \in \mathcal{A}_c$  with witness  $d$ . By (L) and Proposition 1(b),  $d \in \ker M_c$  and  $v_i = -z_i^\top X_i d / z_i^\top z_i$  for  $i \in S_c$ . For  $i \notin S_c$ , (T) at any  $t$  with  $z_i(t) \neq 0$  gives  $v_i = -x_i(t)^\top d / z_i(t)$ , which by Condition S is  $v_i(d)$ . So  $v = v(d)$ .

If  $z_i(t) = 0$  for every  $t$  as well as at every observation, then column  $c$  of subject  $i$ ’s random effects does not enter the likelihood at all,  $v_i$  is unrestricted, and the corresponding coordinate is identified by its prior alone; it plays no part in the ridge.  $\square$

**A sufficient condition for Condition S.** Suppose each column of the fixed-effects design and column  $c$  of the random-effects design are, for subject  $i$ , functions  $f_j(t; w_i)$  and  $g(t; w_i)$  of time and baseline covariates, and that for each  $d$  and scalar  $s$  the function  $\sum_j d_j f_j(\cdot; w_i) + s g(\cdot; w_i)$  lies in a linear space of functions that is determined by its values at subject  $i$ ’s  $n_i$  observation times (for example, polynomials of degree less than  $n_i$ ). Then, for  $i \in S_c$ , (L) at the observation times implies the identity at all  $t$ , which is Condition S for that subject. Subject-constant columns, a linear time column and covariate-by-time interactions satisfy this with  $n_i \geq 2$ .